

# CrohnIPI Diffusion: Class-Balanced Synthetic Augmentation for Crohn's Endoscopy Classification

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## Motivation

Clinical endoscopy datasets are often highly imbalanced, where majority classes dominate the training data. This imbalance makes it difficult for classification models to learn rare Crohn's disease [1] findings such as ulceration, edema, and stenosis.

Our work investigates whether diffusion-generated synthetic images can help improve classification performance under severe class imbalance.

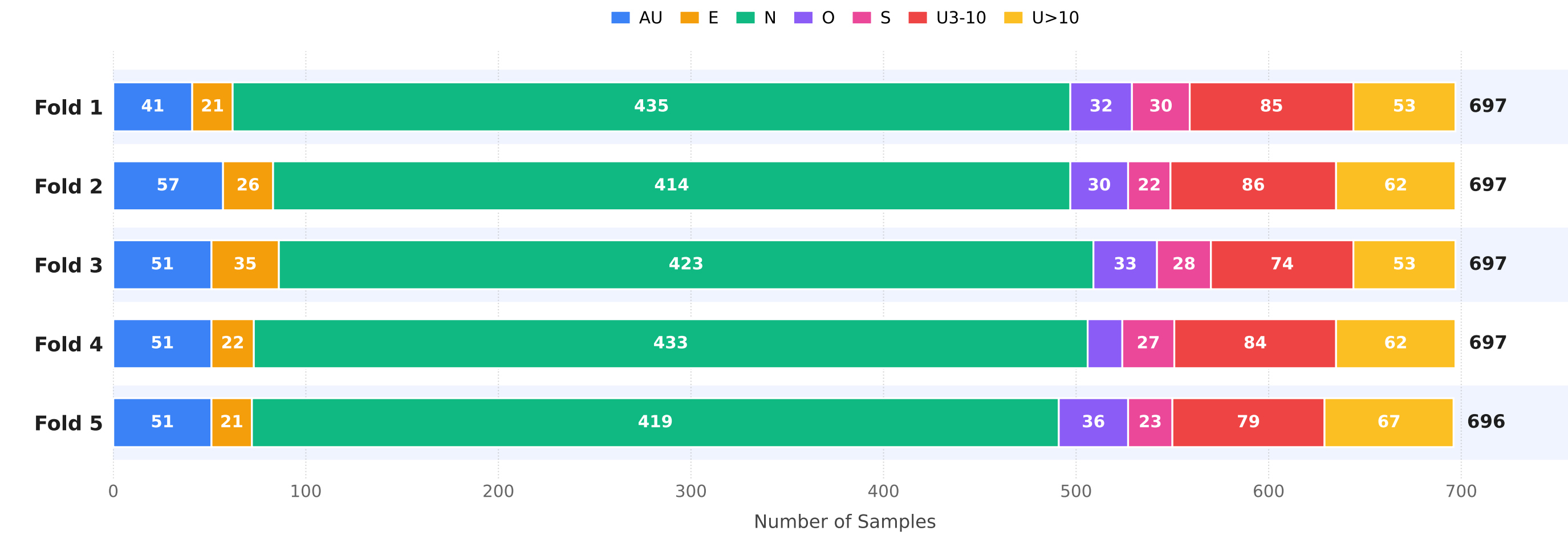


Figure 1. CrohnIPI dataset distribution across five folds. The normal class dominates the dataset, creating a strong class-imbalance problem.

## Dataset

Dataset: CrohnIPI endoscopy image dataset

Classes:

| Class          | Full name                         |
|----------------|-----------------------------------|
| N              | Normal                            |
| E              | Erythema                          |
| O              | Edema                             |
| AU             | Aphthoid Ulceration               |
| S              | Stenosis                          |
| U3-10          | Ulceration between 3 mm and 10 mm |
| U>10 / U_gt_10 | Ulceration greater than 10 mm     |

Evaluation setup: 5-fold cross-validation

Main challenge: Severe class imbalance, where the majority class dominates and rare pathological classes have fewer examples.

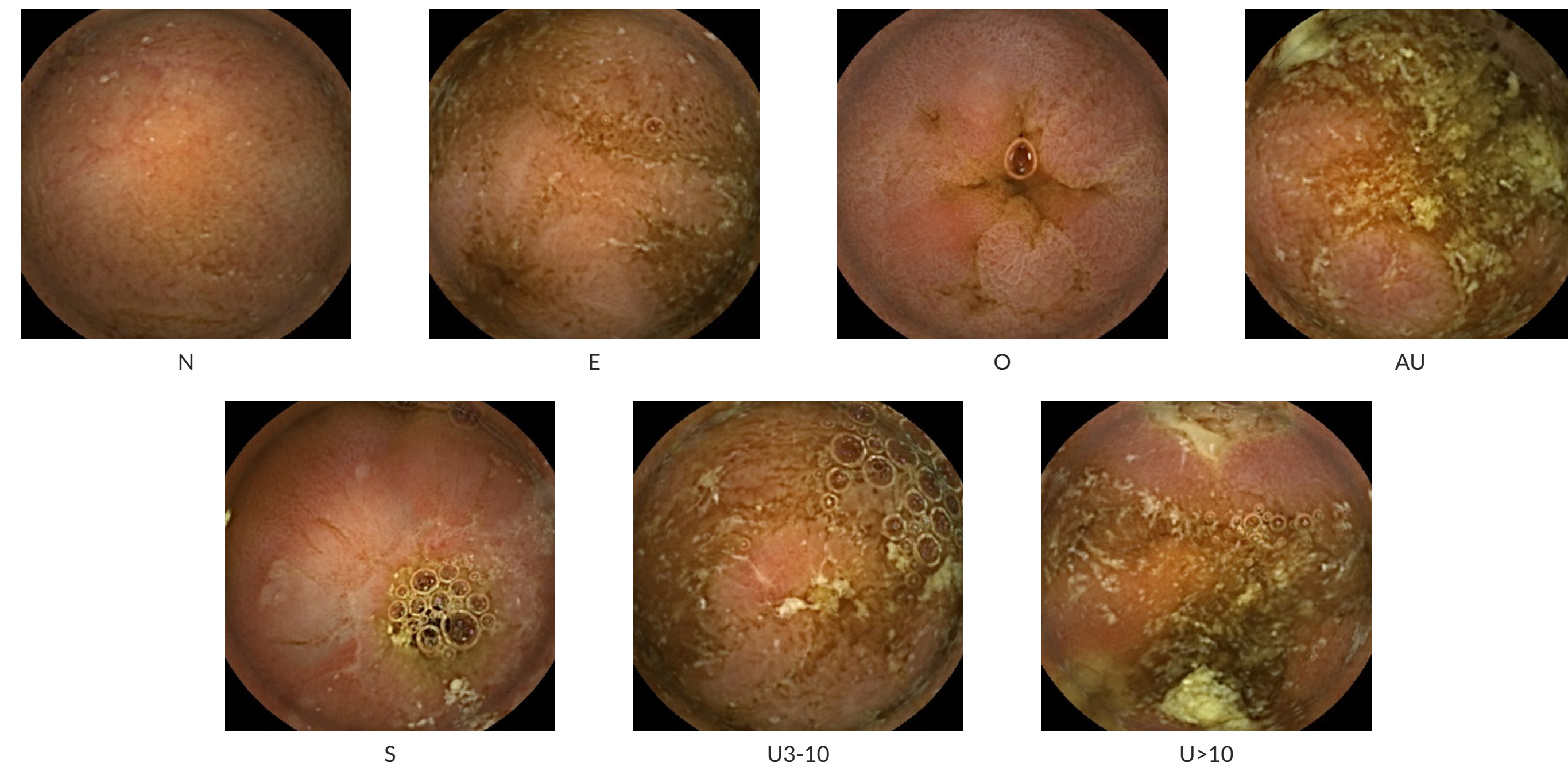


Figure 2. Representative CrohnIPI endoscopy examples from the seven dataset classes.

## Research Question

Key question: To what extent can per-class diffusion-generated synthetic images improve the classification performance of CrohnIPI models under severe class imbalance?

## Proposed Method

We train per-class LoRA diffusion [3] adapters using real CrohnIPI images. Each adapter generates synthetic images for a specific clinical class. The generated images are then passed through multiple filtering steps before being used to train a ConvNeXt-Large [2] classifier.

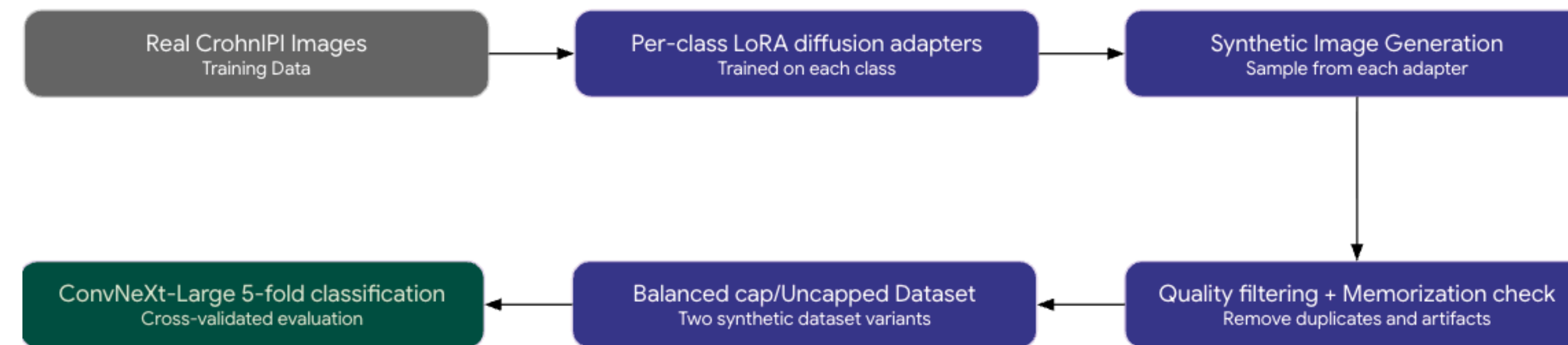


Figure 3. Overview of the proposed synthetic augmentation pipeline using per-class LoRA diffusion adapters and filtered synthetic data for classifier training.

## Synthetic Image Filtering

To reduce low-quality or unsafe synthetic samples, we apply several filtering steps before using generated images for downstream classifier training.

| Filtering Step               | Purpose                                                     |
|------------------------------|-------------------------------------------------------------|
| 1. Artifact detection        | Remove blurry or broken-color images                        |
| 2. Class-consistency check   | Remove images predicted as the wrong class                  |
| 3. DINOv2 memorization check | Detect synthetic images that are too similar to real images |
| 4. Diversity deduplication   | Remove highly similar synthetic samples                     |
| 5. Final filtered set        | Used for downstream classifier training                     |

Table 1. Synthetic image filtering steps used to remove low-quality, inconsistent, memorized, or duplicate generated samples.

Filtering thresholds:

- Laplacian variance < 50
- Channel std < 0.02
- ConvNeXt-Tiny confidence < 0.70
- Pairwise cosine similarity >= 0.95

## Experiment Design

We compare real-only training against standard augmentation, oversampling, and multiple synthetic augmentation settings.

| Condition                 | Description                             |
|---------------------------|-----------------------------------------|
| Real only                 | Baseline using only real images         |
| Real + Augmented          | Standard clinical image augmentation    |
| Oversampling              | Repeated rare-class real images         |
| Synthetic uncapped        | Full per-class LoRA synthetic dataset   |
| Synthetic balanced/capped | Equal synthetic cap per class           |
| Focal loss [4]            | Synthetic data with focal loss [4]      |
| Inverse-frequency CE      | Synthetic data with class-weighted loss |

Table 2. Training conditions used to evaluate whether synthetic diffusion augmentation improves CrohnIPI classification under severe class imbalance.

## Synthetic Dataset Design

Synthetic images were generated for each class using per-class LoRA adapters. We evaluate both uncapped synthetic datasets and balanced/capped synthetic datasets to study whether synthetic class balancing improves classification performance.

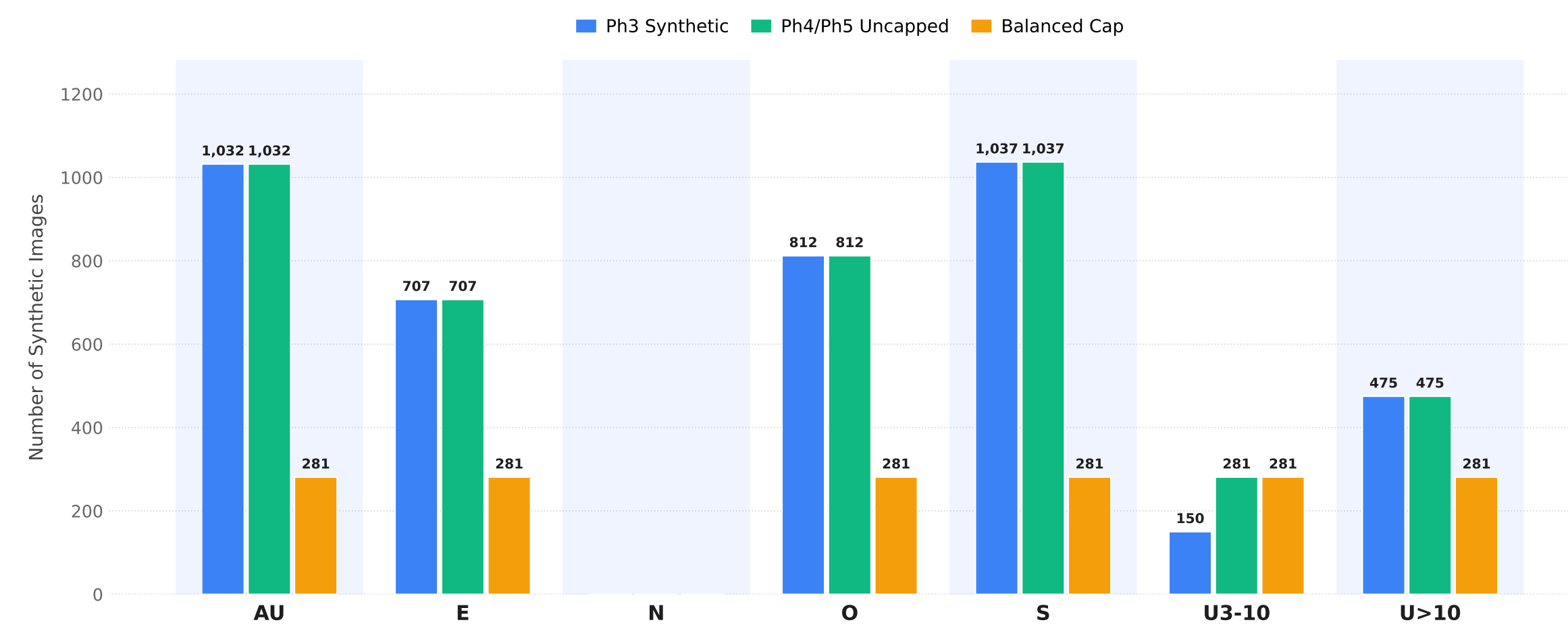


Figure 4. Number of generated synthetic images per class across different synthetic dataset variants.

## Main Results

| Method               | Macro F1      | Balanced Accuracy | Macro AUROC   |
|----------------------|---------------|-------------------|---------------|
| Real only            | 0.7888        | 0.7781            | 0.9709        |
| Real + Augmented     | 0.7973        | 0.7902            | 0.9727        |
| Oversampling         | 0.7786        | 0.7596            | 0.9628        |
| Synthetic            | 0.8085        | 0.7949            | 0.9708        |
| Synthetic + Balanced | 0.8076        | <b>0.8090</b>     | 0.9591        |
| Focal Uncapped       | <b>0.8139</b> | 0.8078            | 0.9708        |
| Focal Balanced       | 0.8077        | 0.7969            | <b>0.9721</b> |

Table 3. Classification performance across real-only, augmentation, oversampling, and synthetic augmentation conditions. Focal Uncapped achieves the best Macro F1.

Key result: Synthetic augmentation improves Macro F1 over the real-only baseline. The best Macro F1 is achieved using the Focal Uncapped synthetic condition.

## Per-Class Recall Analysis

Synthetic augmentation improves recall for several rare classes, but the gains are not uniform across all labels. This suggests that diffusion-based augmentation can help with class imbalance, but class-specific generation quality remains important.

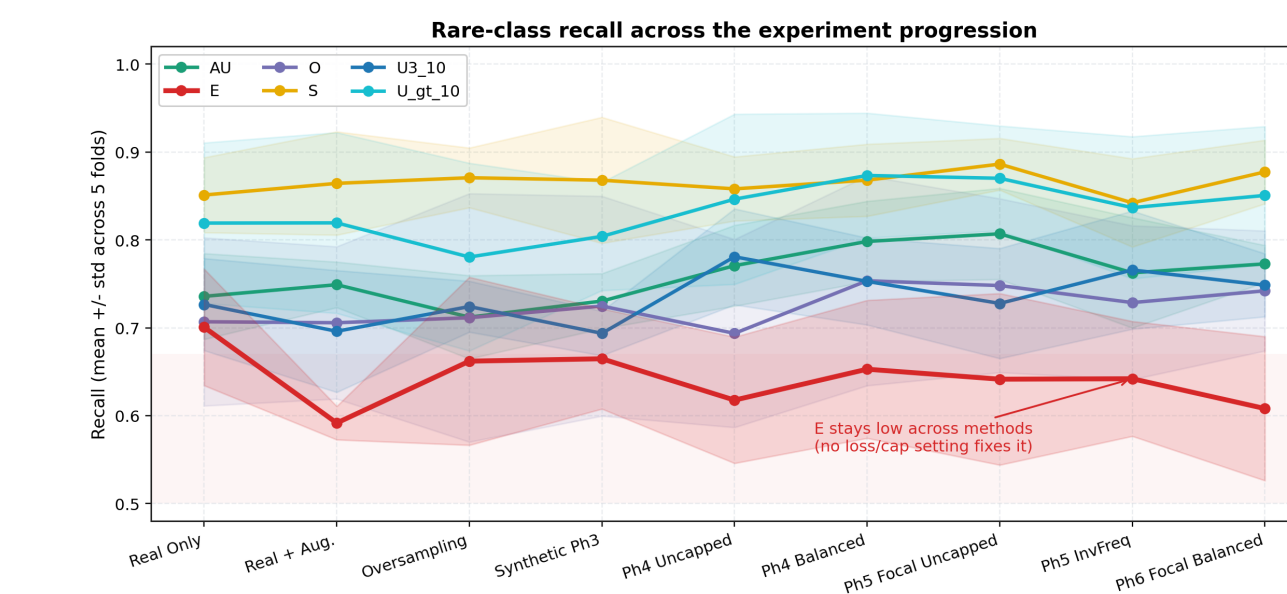


Figure 5a. Rare-class recall across conditions; synthetic augmentation improves several rare classes.

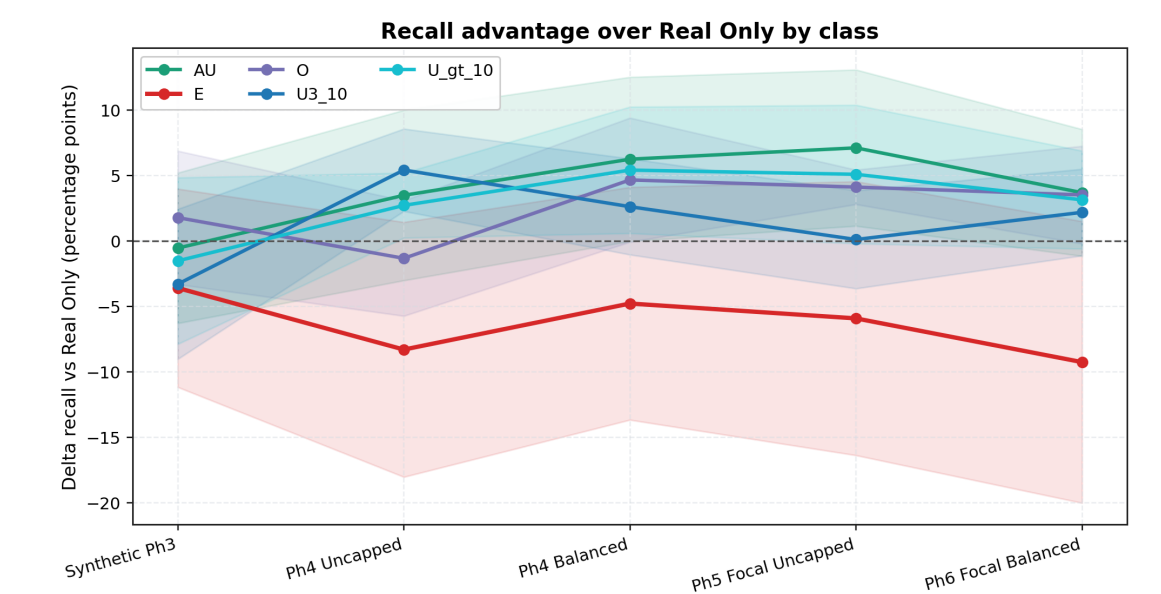


Figure 5b. Recall advantage relative to the real-only baseline.

## Failure Analysis

The current method still struggles with the E class. Failure analysis shows that some E samples are confused with other classes, including the majority class. This indicates that synthetic data alone is not enough; better class-specific generation and expert-guided filtering may be needed.

### Why E is the unresolved class under Ph4 Balanced

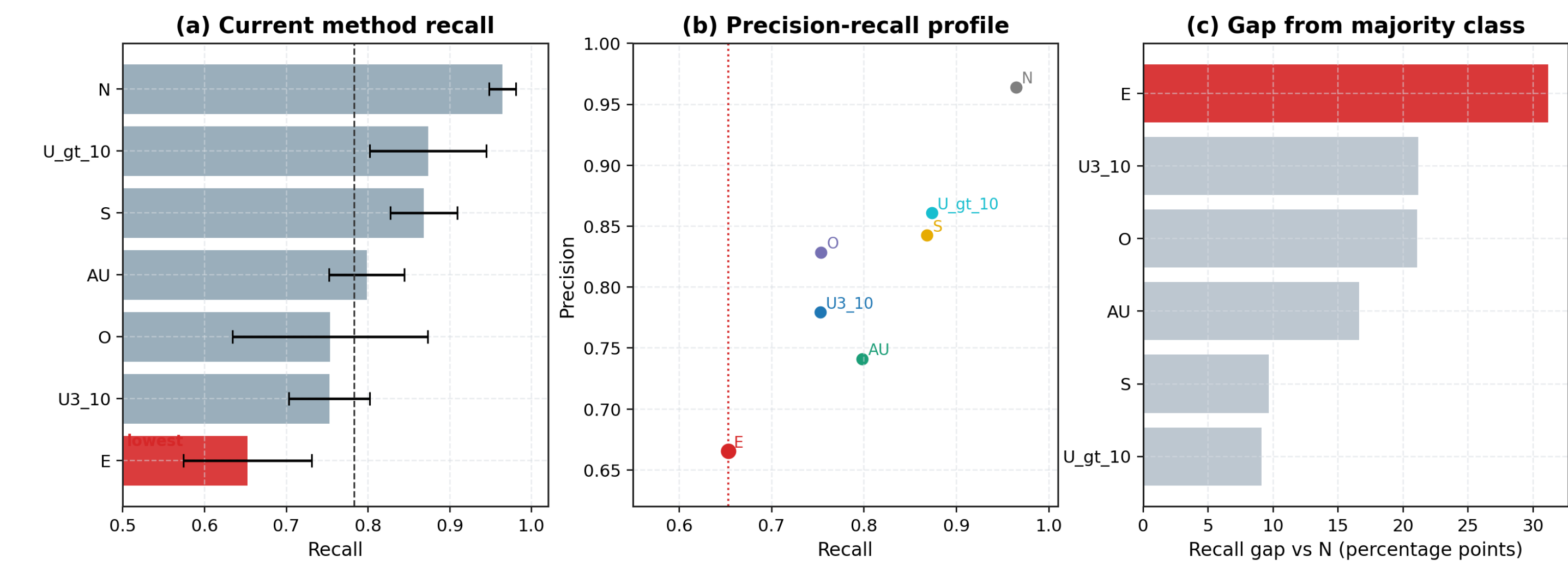


Figure 6. Failure analysis showing that the E class remains difficult under the current synthetic augmentation method.

## Main Takeaways

- Per-class LoRA synthetic data improves class-imbalanced CrohnIPI classification.
- Synthetic augmentation outperforms oversampling; Focal Uncapped gives the best Macro F1.
- Balanced and focal settings improve complementary metrics.
- The E class remains challenging, motivating stronger filtering and expert validation.

### References

- [1] Vallée et al., CrohnIPI, SPIE Medical Imaging, 2020.
- [2] Woo et al., ConvNeXt V2, CVPR, 2023.
- [3] Podell et al., SDXL, ICLR, 2024.
- [4] Mukhoti et al., Calibrating deep neural networks using focal loss, NeurIPS, 2020.